

Business Requirement Document

Route-Based Ride Sharing Platform

Prepared for concept validation and solution planning

This document describes a ride-sharing application similar to Uber in usability, payments, and trip tracking, but different in business concept. Instead of on-demand drivers responding to passenger requests, drivers publish their own planned routes and passengers join rides that match all or part of their journey.

Prepared by: CMMBOT

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1. Executive Summary

The proposed platform is a **planned-route ride sharing application** that helps drivers monetize empty seats during trips they are already planning to make, such as a daily home-to-office commute. Passengers can search for rides based on pickup and destination, then choose an available vehicle that covers the whole route or the largest possible portion of it.

The platform delivers value to three parties. **Drivers** earn money from unused seats, **passengers** travel for less than a taxi ride, and the **platform** earns a commission on each completed trip. The core innovation is route matching based on the driver's pre-defined trip rather than a passenger-initiated taxi request.

2. Business Idea and Product Vision

- Drivers publish one-time or recurring trips they are already planning to take.
- Passengers search for rides matching their pickup and drop-off locations.
- The system returns full-route and partial-route matches.
- Passengers choose the ride that best fits price, overlap distance, convenience, and timing.
- The platform supports cash and card payments and deducts a small commission.

3. How This Differs from Uber

The user experience may feel familiar to Uber, but the operational model is different. Uber is an on-demand taxi marketplace where the driver's route is created around the passenger's trip request. In this platform, the driver is already traveling for personal or routine reasons and simply shares available seats with passengers whose routes overlap.

- **Uber:** Passenger requests a ride first; driver responds to that request.
- **This app:** Driver publishes a planned route first; passenger searches among existing routes.
- **Uber:** Driver is expected to pick up and drop off based on passenger need.
- **This app:** Passenger adapts to the driver's route and may ride only part of the journey.
- **Uber:** Price is typically tied to the passenger's full private trip.

- **This app:** Price is tied to the actual distance traveled inside a shared journey.

4. Example Use Case

A driver travels every weekday from **Rajagiriya to Colombo Fort** at 8:00 AM. The driver uses only one seat in a four-seat car, so three seats are empty. The driver publishes this recurring route in the app.

A passenger wants to travel from **Narahenpita to Bambalapitiya**. The app finds that the driver's route overlaps 80% of the passenger's journey. The passenger books the ride, joins at a pickup point along the route, travels until the driver's route diverges, and pays only for the distance actually traveled.

- Driver benefits by earning extra income on an already planned journey.
- Passenger benefits from a lower cost than a taxi ride.
- Platform benefits by earning a small commission on the completed trip.

5. Business Objectives

- Create a low-cost transport option for passengers.
- Help private drivers earn money from unused seats.
- Encourage shared mobility and reduce empty-seat travel.
- Build a scalable route-matching marketplace for daily and occasional travel.
- Earn recurring revenue through trip commissions and potential subscription or wallet services later.

6. Stakeholders

- **Passengers:** People looking for affordable travel between locations.
- **Drivers:** People already making a trip and willing to share empty seats.
- **Platform Owner:** Business operating the marketplace and earning commission.
- **Administrators:** Staff responsible for verification, support, disputes, and system management.
- **Payment Partners:** Card processors and settlement services.
- **Map and GPS Providers:** Services enabling route calculation and live tracking.

7. User Types

- **Passenger:** Searches, books, pays, tracks rides, and rates drivers.
- **Driver:** Creates routes, manages seats, starts trips, tracks passengers, and receives earnings.
- **Admin:** Verifies drivers, manages pricing/commission rules, handles disputes, and monitors trips.

8. Core Functional Requirements

- User registration, login, and profile management.
- Driver and vehicle verification.
- Trip creation for one-time and recurring routes.
- Passenger search by pickup, drop-off, and time.

- Smart route matching for full and partial trips.
- Seat booking and capacity management.
- Trip start, live tracking, boarding, drop-off, and completion.
- Dynamic fare calculation based on actual distance traveled.
- Cash and card payments.
- Commission calculation and settlement.
- Trip history, receipts, ratings, and support handling.

9. Detailed Functional Requirements - Passenger Module

- Register using phone number, email, or social login.
- Verify account using OTP or email confirmation.
- Save favourite locations such as Home and Office.
- Enter pickup and destination locations.
- Search available rides by date/time.
- View rides in list and map view.
- Compare rides by route match percentage, estimated price, driver rating, ETA, and available seats.
- Book one or more seats based on policy.
- Pay by cash or authorize card payment before trip start.
- Track driver location and ride progress in real time.
- Leave the ride before the driver reaches the final route destination.
- Receive fare receipt based on actual traveled distance.
- Rate and review the driver.
- View booking and travel history.

10. Detailed Functional Requirements - Driver Module

- Register as a driver and complete KYC/document verification.
- Add one or more vehicles with seat capacity and registration details.
- Create one-time or recurring routes.
- Set departure time, available seats, and route schedule.
- Publish a planned trip in advance.
- Start the trip only when the journey actually begins.
- View passenger bookings and pickup points.
- Enable manual approval or instant booking based on platform rules.
- Track occupancy and remaining available seats.
- Mark passenger boarded and dropped off.
- Complete the trip and receive earnings summary.

- View trip history, payouts, and ratings.

11. Detailed Functional Requirements - Admin Module

- Approve or reject driver applications and vehicle documents.
- Manage passengers and drivers.
- View all planned, live, completed, and cancelled trips.
- Configure commission rates and fare rules.
- Set route deviation tolerance and service area rules.
- Handle refunds, complaints, penalties, and fraud cases.
- Generate business reports and analytics.
- Monitor payment and settlement status.

12. Trip Lifecycle

A trip should move through the following states in a predictable way to support booking, billing, and tracking.

- Trip created by driver.
- Trip published and visible for search.
- Passenger books available seat.
- Payment is authorized or payment mode is confirmed.
- Driver starts trip at actual departure time.
- Passenger boards vehicle.
- Passenger rides full or partial route.
- Passenger drops off when route overlap ends or by choice.
- Fare is finalized based on actual distance traveled.
- Trip and payment are completed; ratings become available.

13. Route Matching Requirements

This is the most critical capability of the platform. The system must compare the passenger's requested route with each driver's published route and determine the suitability of the match.

Matching should not depend only on exact pickup and drop-off equality. It must support route overlap, proximity tolerance, and a ranking model so passengers can choose the most useful option.

- Identify rides with full route coverage.
- Identify rides with high overlap such as 50%, 70%, or 90% of the passenger journey.
- Support configurable pickup/drop proximity, such as within 500 meters or 1 kilometer of the route.
- Show route match percentage, estimated ride distance, time savings, and price.
- Rank results by a combination of route overlap, time compatibility, detour impact, price, and driver quality.

14. Example Route Matching Scenarios

- **Scenario A - Full Match:** Passenger route is fully covered by the driver route. Passenger can complete the trip with one driver.
- **Scenario B - Partial Match:** Driver covers only half the passenger route. Passenger may still select the trip if it meaningfully reduces travel cost.
- **Scenario C - High Overlap:** Driver covers 90% of the route. Passenger only needs a short onward journey after getting off.
- **Scenario D - Nearby Match:** Passenger pickup is not exactly on the route but lies within the acceptable pickup radius.

15. Pricing and Fare Rules

- Fare should be based primarily on actual distance traveled by the passenger.
- Time-based or surge-based adjustments may be introduced later but are not mandatory for initial launch.
- The system must support cash and card payments.
- For card payments, the system should pre-authorize before trip start and capture the final amount after actual travel is known.
- For cash payments, the app should record cash collection while still calculating platform commission.
- The app should generate digital receipts and transaction history.

16. Example Fare Calculation

Suppose a driver publishes a 20 km route. A passenger uses only 12 km of that route. If the pricing model is LKR 50 per km and the platform commission is 10%, the total passenger fare is LKR 600.

The driver receives LKR 540 and the platform keeps LKR 60 as commission. If the passenger exits earlier than expected, the fare should be recalculated according to the actual distance traveled.

17. Payment Requirements

- Support card payment, wallet integration, and cash payment.
- Support pre-authorization for card trips before ride start.
- Finalize or adjust payment after actual ride completion.
- Store payment status, refund status, and settlement status.
- Maintain audit logs for payment disputes.

18. Recurring Trip Requirements

Recurring trips are a major differentiator because many users travel along the same route every day or every week.

- Drivers should be able to create recurring daily or weekly routes.
- Driver should define days of week, route, departure time, and seat count.
- Passengers should be able to book single occurrences or potentially repeated bookings in future phases.
- The system should automatically generate future trip instances from the recurring schedule.

19. Booking and Capacity Rules

- A booking must not exceed available seat count.
- The system must prevent overbooking during concurrent bookings.
- Booking status should include requested, confirmed, boarded, completed, cancelled, and no-show states.
- Capacity should update instantly when seats are booked, cancelled, or trip begins.

20. Safety and Trust Requirements

- Driver identity verification and vehicle verification.
- Passenger identity verification for fraud reduction.
- Live trip tracking and trip share link for trusted contacts.
- Emergency/SOS button.
- Ratings and reviews for both passengers and drivers.
- In-app support and dispute reporting.

21. Notifications

- Booking confirmation notifications.
- Driver trip start notifications.
- Driver arrival at pickup point notifications.
- Payment confirmation and receipt notifications.
- Trip completion notifications.
- Cancellation and delay alerts.

22. Reporting and Analytics

- Trips created vs trips completed.
- Average route overlap percentage.
- Top recurring routes.
- Revenue and commission reports.
- Driver earnings reports.
- Passenger booking trends.
- Cancellation and fraud reports.

23. Non-Functional Requirements

- **Performance:** Fast route search and responsive live tracking.
- **Scalability:** Support growing users, trips, and concurrent live sessions.

- **Security:** Protect payment and personal data with encryption and secure authentication.
- **Reliability:** Preserve trip states correctly during network loss or app restart.
- **Usability:** Keep passenger booking flow simple and map-based.
- **Auditability:** Track trip, payment, and settlement events for dispute resolution.

24. Assumptions

- Drivers are ordinary users making planned trips, not full-time taxi operators.
- Passengers are willing to accept partial route coverage for lower cost.
- Map and GPS services are available and accurate enough for route matching.
- The business can legally operate shared rides in the target market subject to transport regulations.

25. Risks and Complex Areas

- Designing accurate route matching and ranking logic.
- Handling actual-distance fare calculation when passengers exit early.
- Real-time tracking stability and battery usage.
- Payment settlement across card, cash, and commission models.
- Managing safety, trust, and fraud controls.
- Supporting recurring schedules without creating booking conflicts.

26. Recommended MVP Scope

- Passenger and driver registration/login.
- Driver route publication for one-time and recurring trips.
- Passenger route search and route-overlap results.
- Booking and seat management.
- Trip start and live tracking.
- Cash and one card payment method.
- Fare calculation based on actual traveled distance.
- Commission and basic admin console.
- Ratings, receipts, and trip history.

27. Future Enhancements

- Corporate commute programs.
- Subscription-based regular commuter plans.
- Shared ride bundles and monthly passes.
- Wallet, promotions, referral system, and loyalty program.
- AI-based ride recommendations and optimal pickup point suggestions.

- Multi-leg journey support using more than one driver.

28. Final Requirement Statement

The proposed solution is a route-based ride-sharing platform where drivers publish planned trips and passengers search for rides that fully or partially match their travel needs. The platform must support one-time and recurring trips, smart route matching, live tracking, seat booking, actual-distance fare calculation, cash and card payments, ratings, safety features, and platform commission management. The goal is to offer a lower-cost transport option while helping drivers earn from unused vehicle capacity.